MILITARY SPECIFICATION

TRANSMITTER, T-827( )/URT

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Transmitter (Exciter), T-827( )/URT and associated accessories, hereinafter referred to as the equipment. The equipment is a digitally tuned transmitter (exciter) intended for general purpose voice, continuous wave (CW), radio teletype and data communications, which operates in the 2.0 megahertz (MHz) to 30.0 MHz frequency range and provides transmission of single sideband (SSB), (selectable upper sideband (USB), lower sideband (LSB) and independent sideband (ISB), CW amplitude modulation (AM compatible), single and multi-channel teletypewriter and Naval Tactical Data System (NTDS) Link 11 signals.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-17/84	Cables, Radio Frequency, Flexible, Coaxial, 50 Ohms, M17/084-RG223
MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-C-915	Cable and Cord Electrical, For Shipboard Use, General Specification For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-W-16878/4	Wire, Electrical, Polytetrofluoroethylene (PTFE) Insulated, 200 Deg. C, 600 Volts, Extruded Insulation

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging And Packing Of
MIL-F-19207	Fuseholders, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For
MIL-C-22954	Control, Radio Set C-1138( )/UR
MIL-T-24038	Transmitter, Radio AN/URT-24( )
MIL-T-28706	Transmitter, Radio AN/URT-23( )
MIL-M-38510	Microcircuit, General Specification For
MIL-C-39012	Connectors, Coaxial, Radio Frequency, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188-114	Electrical Characteristics Of Digital Interface Circuits
MIL-STD-188-203-1	Subsystem Design And Engineering Standards For Tactical Digital Information Links (TADIL) A
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-965	Parts Control Program
DOD-STD-1399 Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MIL-STD-1633	Interface Standard For Shipboard Emission Monitor-Control Set, AN/SSQ-82(V) MUTE System

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2.1.2 Other Government drawings. The following other Government drawings, form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of the solicitation.

DRAWINGS

MILITARY

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

RE-C2687412	Electrical Power Cable Assembly CX-1846( )/U

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

RE-F-2688036	Handset H-169( )/U

(Copies of specifications, standards, handbooks and drawing required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77	Racks, Panels And Associated Equipment

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3). The sample(s) shall be suitable for complete evaluation of mechanical and electrical form, design and performance. The sample shall be of final mechanical and electrical form, employ standard or nonstandard (when approved by the procuring activity) parts, and be completely representative of final production equipment.

3.1.2 Parts, materials and processes. Selection of parts, materials and processes shall be in accordance with the Parts materials and processes paragraph of MIL-E-16400, except as specified in 3.1.2.1 through 3.1.2.7. Particular attention shall be given to the acceptable and unacceptable materials of MIL-E-16400.

3.1.2.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

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3.1.2.2 Prior parts approval. The equipment furnished in accordance with this specification shall employ standard parts or approved nonstandard parts. Nonstandard part approvals previously granted under earlier contracts for equipments represented by Government-furnished equipment (GFE) shall not be construed as being extended to equipments furnished in accordance with this specification.

3.1.2.3 Individual selection. The performance of the equipment shall not be dependent on the selection of individuals parts, unless such selection is specifically approved by the procuring activity.

3.1.2.4 Electron tubes. Electron tubes shall be selected from among approved types in accordance with the Tubes, electron paragraph of MIL-E-16400. Electron tubes shall be limited to applications corresponding to circuits of the GFE.

3.1.2.5 Microelectronic devices. Selection and application of microelectronic devices shall conform to MIL-STD-454, Requirement 64, requiring MIL-M-38510, Class B devices. When deviations from these requirements are necessitated due to availability or cost considerations, hermetically sealed devices shall be inspected and tested in accordance with the Other microcircuits paragraph of MIL-STD-454, Requirement 64.

3.1.2.6 Semiconductor devices. Semiconductor devices shall be made of silicon.

3.1.2.7 Type BNC and Type N radio frequency (RF) connectors. Where Type BNC and Type NRF coaxial connectors are specified herein, connectors shall be in accordance with MIL-C-39012.

3.1.3 Production equipment. Production equipment shall be identical to the equipment that has been subjected to and passed the first article examinations and tests (see 3.1.1 and 4.3).

3.1.4 Operational requirements. Each unit shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in Naval ship and shore communications service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

3.2 GFE. GFE, if any, shall be as specified, (see 6.2.1 and 6.4).

3.3 Interchangeability. Interchangeability requirements shall be as specified (see 6.2.1).

3.4 Composition. The equipment shall consist of one each of the items specified in a through c:

a. Transmitter (Exciter), T-827 ( )/URT
b. Handset, H-169( )/U, in accordance with drawing (DWG) RE-F-2688036 including cord and plug assembly, CX-1846( )/U, in accordance with DWG RE-C2687412
c. Set of mating connectors including cable clamps for equipment system connection as specified in 1 through 8:
    1. USB audio
    2. LSB audio
    3. Radio teletypewriter (RATT)
    4. Data modem audio and keyline
    5. Alternating current (AC) power
    6. RF output
    7. Frequency standard input
    8. Frequency standard output

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3.4.1 Accessories. In addition to the basic equipment specified in 3.4, the accessory items specified in a and b shall be furnished in the quantities specified (see 6.2.1).

a. Equipment plug-in assembly extension cable (see 3.15.33.1) as specified in 1 through 6:

1. RF amplifier assembly cables (two each)
2. Frequency standard assembly cable (one each)
3. Translator and synthesizer assembly cables (two each)
4. Transmit intermediate frequency (IF) assembly cable (one each)
5. Transmit mode selector assembly cable (one each)
6. RATT tone generator assembly cable (one each)

b. Set of printed circuit board (PCB) extender cards, one for each plug-in PCB in the equipment (see 3.15.1.33.2).

3.5 Frequency range. The frequency range (tuning) of the equipment shall be from 2.0 MHz to 30.0 MHz. Each equipment shall be capable of adjustment to any frequency from 2.0 MHz to 29.9999 MHz in 100 hertz (Hz) steps.

3.5.1 Method of tuning. Frequency selection over the complete frequency range of the equipment shall be accomplished in steps of 100 Hz where changing the setting of the digital tuning knobs is all that is required to tune the equipment (see 3.15.1.6).

3.6 Primary power source. The equipment shall be designed to operate from electrical power having characteristics as specified for Type I in DoD-STD-1399, Section 300, and as specified in 3.6.1 through 3.6.3. Each equipment shall be capable of operating and providing rated performance from a primary power source of 115 volts alternating current (VAC) ±10 percent 48 Hz to 420 Hz, single phase, with no degradation of performance by power line voltage variation (relative to the nominal) of ±10 percent or above specified power line frequency variations, with either of these variations acting separately or in combination. The primary of the AC power transformers shall be equipped with connections to the nominal 115 volt (V) tap. The total power consumption at 115 V, 60 Hz or 400 Hz shall not exceed 70 watts (W).

3.6.1 Transient protection. The equipment shall operate without damage or permanent change in characteristics due to transient voltage, transient frequency and power interruption test of MIL-E-16400 and the 2500 V spike test of DoD-STD-1399, Section 300. Protection shall be included against damage due to transients produced by operation of relays or tuning motors within the equipment.

3.6.2 Electrical overload protection. Electrical overload protection shall be in accordance with MIL-E-16400, except that each line of the AC power input shall be protected by fuses.

3.6.3 Leakage current. The equipment leakage current measured in accordance with the procedures shall not exceed 5 milliamperes (mA) AC or direct current (DC).

3.7 Operational requirements. Operational requirements shall be as specified in 3.7.1 and 3.7.2.

3.7.1 Type of emission. Each equipment shall provide for the transmission of the emissions specified in a through j:

a. A1 telegraphy without the use of modulating audio frequency CW
b. A3a telephony, SSB, reduced carrier
c. A3b telephony, two ISBs, reduced carrier
d. A3h telephony, SSB, full carrier
e. A3j telephony, SSB, suppressed carrier
f. A9b two ISBs, combination of telephony and telegraphy ISB/RATT
g. F2j single channel voice frequency telegraphy RATT
h. F7j multi-channel voice frequency telegraphy RATT
i. F9 differentially coherent phase shift keying (DPSK): tactical digital information link (TADIL) A as specified in MIL-STD-188-203-1, double sideband (DSB)
j. F9j: TADIL A as specified in MIL-STD-188-203-1, SSB

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3.7.2 Modes of operation. The equipment shall be capable of transmitting the emissions specified in 3.7.1 by selection of one of the modes of operation specified in a through g:

a. LSB
b. USB
c. ISB
d. AM (compatible)
e. CW telegraphy
f. Single or multi-channel telegraphy RATT
g. LSB audio and USB single channel telegraphy ISB/RATT

3.8 Equipment arrangement. Normal installation arrangement shall include mounting the equipment as part of an AN/URT-23( ) or AN/URT-24( ) configuration in accordance with MIL-T-28706 and MIL-T-24038.

3.9 Radio set system. The contractor shall be responsible for demonstrating, subject to availability of equipments, the operational compatibility of the T-827( )/URT with AN/URT-23( ) or AN/URT-24( ) or both. Equipment provisions for external cabling to the standard Navy shipboard radio remote control systems through the AN/URT-23( ) or AN/URT-24( ) shall be required. Navy cables shall be in accordance with MIL-C-915. Mating connectors for all external cabling which require manufacture by the installing activity shall be furnished with each equipment (see 3.4c).

3.9.1 Radio remote control system. The equipment shall contain compatible provisions for remote control of the equipment in the standard Navy radio remote control system (not frequency control) using a maximum of three radio set Control Units, C-1138( )/UR, in accordance with MIL-C-22954. Continuous simultaneous key-down operation from only two C-1138( )/UR units shall be required. Start and stop provisions are not required with use of C-1138( )/UR control units.

3.9.2 Multiple unit for transmission elimination (MUTE). The T-827( )/URT shall be capable of interfacing with the MUTE system in accordance with MIL-STD-1633.

3.10 Dimensions and weight. The dimensions and weight of the equipment shall be as specified in TABLE I.

TABLE I. Dimensions and weight.

Unit	Width cm 1/ (in.) 2/	Depth 3/ cm (in.)	Height cm (in.)	Weight kg 4/ (lb) 5/
Equipment	44.13 (17.375)	47.0 (18.50)	17.75 (7.0)	34.0 (75.0)

1/ Centimeters
2/ Inches
3/ Exclusive of rear connectors
4/ Kilograms
5/ Pounds

3.11 Environmental service conditions. The equipment shall be capable of continuous, reliable operation, within the performance limits specified in 3.16, under the environmental conditions of MIL-E-16400 specified in a through g and in 3.16. The equipment shall operate within tolerances throughout the corresponding test specified in Section 4 without alignment or adjustment, other than external controls normally employed for operation of the equipment.

a. Operating temperature range (Range 4)
b. Nonoperating temperature range (Range 4)
c. Humidity ($\leq$ 95 percent)
d. Salt fog (spray) (sheltered)
e. Shock
f. Vibration
g. Inclination (60 degrees)

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3.12 Orientation. The equipment shall be capable of storage in any orientation.

3.13 Quantitative reliability requirement. The upper test mean-time between-failure (MBTF), (θ_0 as defined in MIL-STD-781) shall be 2000 hours. The lower test MTBF (θ_1 as defined in MIL-STD-781) shall be 1000 hours.

3.14 Basic conditions. The basic conditions specified in 3.14.1 through 3.14.1.7 shall apply to the performance requirements specified in 3.16 except as modified by a particular test or test procedure specified in this specification. The conditions are specifically applicable to the measurement of the output power, spurious responses, intermodulation (IM) distortion, bandwidth and carrier suppression in the equipment.

3.14.1 Standard test conditions. The standard test conditions shall be as specified in 3.14.1.1 through 3.14.1.7.

3.14.1.1 Primary power. The primary power test condition shall be 115 VAC, 60 Hz, single-phase.

3.14.1.2 Ambient temperature. The ambient temperature shall be between 20° Celsius (C) and 30°C.

3.14.1.3 Relative humidity. The relative humidity shall be less than 95 percent.

3.14.1.4 External and internal frequency standard switch. The frequency standard switch shall be adjusted to the internal or compare position.

3.14.1.5 Internal frequency standard. The internal frequency standard adjustment control shall be adjusted as required to ensure correct setting of the output frequency.

3.14.1.6 Equipment RF output. Equipment RF output shall be measured with a 50-ohm resistive load connected to the RF output terminal through a 1.5 meter (m) (5 foot) length of RG-223 cable conforming to MIL-C-17/84.

3.14.1.7 Audio frequency local input. Local input performance shall be demonstrated using the H-342( )/U Handset or an equivalent device.

3.15 Mechanical design and construction. Mechanical design and construction shall conform to the Mechanical design and construction paragraph of MIL-E-16400 to the extent of being compatible with GFE.

3.15.1 General requirements. General requirements shall be in accordance with 3.15.1.1 through 3.15.1.3.5.

3.15.1.1 Construction. Construction shall conform to the Construction paragraph of MIL-E-16400. The equipment shall consist basically of a chassis, a front panel, a case and group of replaceable plug-in assemblies. The chassis shall include the power supply, receptacles for insertion of the plug-in assemblies, and mechanical hardware associated with frequency selection. The front panel shall contain the required control functions and shall be securely mounted to the chassis. Two handles shall be provided, one mounted on each side of the front panel, and each capable of supporting the entire weight of the unit. All sharp corners and edges on cases, chassis, panels, and brackets shall be eliminated.

3.15.1.2 Case. The case shall be made of aluminum, and shall be of molded construction. A retractable cable and a chassis mating receptacle shall be provided to enable operation of the equipment when the frame is withdrawn from the case and oriented +90 degrees from the normal horizontal position. Additional accessory cables shall not be required for operation when the equipment is withdrawn from the case. The retractable cable shall be secured to the chassis by a cable clamp and installed so that the cable has no abrasion points and does not jam when the chassis is inserted in or withdrawn from the case.

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3.15.1.2.1 Rack mounting. The equipment shall be designed for installation in a standard rack manufactured in accordance with EIA RS-310-C-77. This type of mounting is limited to shore station application. The case of the equipments shall be designed and constructed so that when mounting brackets are installed on the sides of the case, and the equipment is installed in the rack and not subject to shock or vibration, no other support or reinforcing members will be required to preclude deformation of any part of the equipment.

3.15.1.3 Enclosure. The enclosure for the equipment shall be in accordance with the Enclosures paragraph of MIL-E-16400. The degree of enclosure of the equipments shall be dripproof (45 degrees).

3.15.1.3.1 External connections. External connections shall be the same as in the GFE. Provision shall be made for accepting data mode audio from the added modem connector, the equipment switchboard connector on an associated AM-3924 linear power amplifier, or both, as installation options. Remote audio input (connectors J5 and J6 on the GFE) shall be utilized for normal (nondata) audio only.

3.15.1.4 Bonding and grounding. The equipment shall conform to the Bonding and grounding paragraph of MIL-E-16400, except as specified in 3.15.1.4.1 and 3.15.1.4.2.

3.15.1.4.1 Case bonding. Case bonding and grounding shall be accomplished by preparing mounting surfaces to provide metal contact between the case and mounting bracket, and between the mounting bracket and shock mount. The area around each threaded insert of the case used to secure the mounting bracket to the case shall be free of paint and chemically treated to preserve the metal and provide a ground. Mounting brackets and the shock mount used with the equipment shall be similarly treated to ensure bonding and grounding.

3.15.1.4.2 Grounding stud. The stud shall be a pressed-in captive insert made integral with the case and constructed of austenitic corrosion-resistant steel. The captive insert shall be flush with the outside surface of the case and shall contain a blind threaded hole to prevent the entrance of mositure into the case. A 25.4 millimeter (mm) (1-in.) diameter area around the threaded insert shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.15.1.5 Corrosion protection, corrosion-resisting treatments. Corrosion protection and corrosion-resisting treatments shall be in accordance with the Corrosion protection/corrosion-resisting treatments paragraph of MIL-E-16400.

3.15.1.5.1 Painting. Painting shall be in accordance with the Painting paragraph of MIL-E-16400, except that the interior surfaces of each metal equipment enclosure shall be painted a flat black to improve thermal efficiency.

3.15.1.6 Frequency selection system. A digital selection system shall be provided. Manually operated knobs on the front panel shall effectively control the frequency and the display of frequency to which the equipment is tuned. Separate controls shall be provided for the 10 MHz, 1 MHz, 100 kilohertz (kHz), 10 kHz, 1kHz and 100 Hz frequency selection, and each control shall cover its range in less than one rotation of the knob.

3.15.1.6.1 Mechanical backlash. The backlash in the mechanical tuning arrangement shall not abrogate the requirements for frequency accuracy and stability. End stops shall not be provided, except for the 1 MHz, 10 MHz and 100 Hz controls. Each digit selected shall include mechanical detent provisions.

3.15.1.6.2 Knob sequence. The sequence of positioning the frequency selector knobs or the manner of speed in which the knobs are rotated shall not affect performance requirements specified in 3.16. Positioning of the frequency selector knobs more than once shall not be necessary in obtaining the desired frequency.

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3.15.1.7 Fastener hardware. Fastener hardware shall conform to the Fastener hardware paragraph of MIL-E-16400, except as otherwise specified herein. All screws, nuts, bolts, and other fastener hardware shall be of austenitic corrosion-resistant steel, except in those applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on equipment front panels and electronic plug-in assemblies. Cadmium plating is not acceptable on the captive-type bolts used on the equipment front panels and electronic plug-in assemblies.

3.15.1.8 Controls, indicators and panel layouts. Controls, indicators and panel layouts, including lights, receptacles, switches, meters (including functions) and fuseholders, shall be the same as furnished in the GFE, unless otherwise specified in 3.15.1.8.1. A dimmer control for the equipment panel lights is not required. The meter scale may be modified to improve readability.

3.15.1.8.1 Mode selector switch. The mode selector switch shall function in a manner which requires a deliberate action in addition to rotation to accomplish power turnoff, which would result in shutdown of the frequency standard. Design of this control shall be included as part of the panel layout, if required. Design shall prevent cross coupling of keyline and audio circuits in all modes of operation. The mode selector switch assembly shall be so designed as to permit replacement as a plug-in assembly.

3.15.1.9 Electrical and RF connectors. Electrical and RF connectors shall be provided in accordance with Connectors, electrical and Connectors, radio frequency paragraphs of MIL-E-16400. All non RF connectors shall be of the solder-lug type; crimped pin types shall not be used. Mating connectors for non RF connectors shall be provided with metal cable clamps which accomodate standard shipboard cables. All multipurpose connectors shall have 10 percent spare pins to the extent of compatibility with the GFE.

3.15.1.9.1 RF input and output (I/O) connectors. The coaxial radio frequency connectors for the external frequency standard input shall be Type N. The coaxial radio frequency connector for the internal frequency standard output and the exciter output shall be Type BNC. The receptacles shall be mounted at the rear of the equipment case and shall be suitably labeled. The center pin of the Type N connectors shall be made captive in all directions.

3.15.1.9.2 Assembly (module) connectors. RF inserts, if used, shall be constrained from nonperpendicular movement to ensure proper mating.

3.15.1.10 Time meter. A time meter is not required.

3.15.1.11 Locking devices. Locking devices on external operating controls are not required unless otherwise specified herein.

3.15.1.12 Panel lights. Panel lights shall be easily replaceable from the front panel without requiring removal of the frame from its case.

3.15.1.13 Motors. Motors, used in the equipment, shall conform to the Motors, dynamotors, rotary power converters, and motor-generators paragraph of MIL-E-16400 to the extent of compatibility with the GFE.

3.15.1.14 Fuses. Fuses shall be in accordance with the Fuses and fuseholders paragraph of MIL-E-16400, except that extra fuses are not required. Fuses shall be mounted on the front panel in blown fuse indicating fuseholders which conform to MIL-F-19207. Spare fuseholders are not required.

3.15.1.15 Filters. Filters used in the sideband selection process shall be electromechanical or crystal filter type.

3.15.1.16 Head telephones and plugs. Head telephones and telephone plugs are not required to be furnished by the contractor.

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3.15.1.17 Hand key and plugs. Hand keys (CW) and plugs are not required to be furnished by the contractor.

3.15.1.18 Assembly identification. All assemblies to which the interchangeability requirements apply (see 3.3) shall be provided with the assigned National Stock Number, Manufacturer's Part Number (including revision indicator), Federal Supply Code Number, and serial number. This information shall be inscribed on the baseplate of the assembly.

3.15.1.18.1 Dust covers. The dust covers of all assemblies employing PCBs shall indicate the location of all test points and all adjustments accessible through the dust cover.

3.15.1.19 Accessibility. In addition to the requirements of the Accessibility paragraph of MIL-E-16400, the circuit mounting boards, components, and tubes shall be easily accessible for maintenance, repair or replacement. The requirement shall be satisfied to the extent of being compatible with the GFE. Components which require adjustment prior to start of operation shall be accessible without removal of dust covers. All new printed wiring boards shall be removable without the unsoldering of any leads.

3.15.1.20 Test points. Test points shall be provided and located to permit the isolation of malfunctioning parts or subassemblies within major units or assemblies without the unsoldering of interconnections or wiring leads. Test points shall be identical to the test points in the GFE for existing circuitry.

3.15.1.21 Circuit alignment. Resonant circuits shall be provided with adjustable trimmers, or equally effective means, to permit accurate alignment. The trimmers shall provide sufficient means of adjustments to compensate for the effects of normal aging of electron tubes or other parts.

3.15.1.21.1 Bandpass filters. Composite bandpass filters may be furnished as sealed units without trimmers, provided the I/O sections do not require adjustment.

3.15.1.22 Factory adjustment controls. The design of the equipment shall include factory or sealed adjustment controls, unless specifically approved by the procuring activity.

3.15.1.23 Shock, vibration, and inclination. Shock, vibration and inclination shall be in accordance with 3.15.1.23.1 through 3.15.1.23.5.

3.15.1.23.1 Shock. The equipment shall be capable of withstanding the shock test for Grade A, deck-mounted, Class II, lightweight, Type A equipment as specified in MIL-S-901.

3.15.1.23.2 Vibration. The equipment shall be capable of withstanding the Type I vibration test of MIL-STD-167-1. During vibration testing, the equipment shall conform to 3.16.4.1. The identification plates requirements specified in MIL-STD-167-1 are not applicable.

3.15.1.23.3 Inclination. The equipment shall conform to the inclination requirements of 3.11 and the Inclination paragraph of MIL-E-16400.

3.15.1.23.4 Shock, vibration and inclination distortion. The shock and vibration mounting assembly shall show no deformation, bottoming or leaning on completion of the shock vibration, or inclination tests.

3.15.1.23.5 Test configuration. The equipment shall be mounted in the AN/URT-23 configuration to determine compliance with the shock, vibration and inclination requirements.

3.15.1.24 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the requirements of Printed wiring paragraph of MIL-E-16400, to the extent of compatibility with the GFE.

3.15.1.25 Welding. Welding shall conform to the requirements of the Welding, structural paragraph of MIL-E-16400.

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3.15.1.26 Attachment of wire and leads. Attachment of wires and leads shall conform to the Attachment of wires and leads paragraph of MIL-E-16400.

3.15.1.27 Internal wiring and cabling. Internal wiring and cabling shall conform to the requirements of the Internal wiring and cabling paragraph of MIL-E-16400. All wiring shall be tetrafluoroethylene insulated. Hook-up wire shall conform to MIL-W-16878/4.

3.15.1.28 Conductors. Conductors shall be in accordance with MIL-E-16400, except that the number and purpose of conductors shall be the same as those used in the GFE.

3.15.1.28.1 Conductor identification. Identification of conductors shall conform to the requirements of the Conductor identification paragraph of MIL-E-16400. The method used for identification shall be consistent throughout the equipment.

3.15.1.29 Identification and marking. Identification and marking of units, assemblies, electrical and moving mechanical parts, and mechanical parts which require removal for normal maintenance, shall be in accordance with the Identification and marking paragraph of MIL-E-16400. Hertz or Hz shall be used for all indication of frequency on hardware and associated software. Reference designators shall be the same as used in the GFE to the maximum extent practicable.

3.15.1.30 Part replacement. Part replacement shall conform to the Part replacement paragraph of MIL-E-16400.

3.15.1.31 Quartz crystal units. Quartz crystal units shall conform to the Quartz crystal units paragraph of MIL-E-16400.

3.15.1.32 Gears and cams. Gears and cams shall conform to the Gears and cams paragraph of MIL-E-16400, except that only metal gears shall be used.

3.15.1.33 Maintenance accessories. Extender cables or extender cards, as applicable, shall be furnished as required (see 6.2.1). Electronic assemblies and PCBs that plug-in to chassis or mother board receptacles, and on which the test points or components are not available for fault isolation, require an extender cable or extender card, as applicable, for fault isolation.

3.15.1.33.1 Extender cables. Extender cables for the assemblies specified in 3.4.1a, and for new assemblies, shall be 610 mm (24 in.) long.

3.15.1.33.2 PCB extender cards. The extender cards shall be of sufficient height to permit access to all test points and components on the PCB for dynamic fault isolation. The extender cards shall conform to the Printed wiring paragraph of MIL-E-16400.

3.15.1.34 Thermal design. Thermal design shall be in accordance with the Thermal design paragraph of MIL-E-16400 to the extent of compatibility with the GFE.

3.15.1.35 Frequency standard assembly construction maintenance considerations. Construction of the frequency standard assembly shall permit maintenance or replacements, or both of a and b:

a. Oscillator and oven control circuits
b. Oven case, as separate plug-in subassemblies

The metal oven case subassembly components (except the heater winding) shall be readily removeable for replacement. No potting or molding shall be employed to secure replaceable components to the oven case. Nylon screws shall not be used.

3.15.2 Human engineering. The contractor shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein except that equipment shall be designed for use by the 5th percentile, female, to the 95th percentile, male.

3.16 Performance. Performance shall be in accordance with 3.16.1 through 3.16.28.

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3.16.1 Power output. The equipment shall be capable of transmitting 250 milliwatts (mW) peak effective power (PEP) in USB or LSB mode, 250 mW PEP total in ISB mode, 125 mW carrier in CW and RATT mode, and 62.5 mW carrier in AM mode, into a 50-ohm load. These outputs shall be available on a continuously keyed basis, without harm to circuitry, parts and tubes conforming to the technical requirements herein, throughout the frequency range specified in 3.5 and into a 50-ohm resistive load.

3.16.1.1 Maximum RF output. An output of 250 mW PEP shall be produced before limiting occurs.

3.16.2 Power output leveling. Power output level change due only to change in the selected output frequency shall be not greater than 10 decibel (dB) (total) within any MHz band or greater than 15 dB (total) over the entire frequency range specified in 3.5. This requirement shall conform to the average power control (APC) voltage fixed at a level which prevents output limiting at the frequency of highest output.

3.16.2.1 Sideband output equalization. Overall transmit gain shall be adjusted so that difference in output power between the two sidebands in the ISB mode, or when switching from USB to LSB, shall be less than 3 dB (total). Equalization shall be independent of audio gain control circuitry.

3.16.3 Modes of operation. Modes of operation shall be as specified in 3.16.3.1 through 3.16.3.6.1.

3.16.3.1 CW(A1). CW operation shall be provided on the carrier frequency. A CW hold circuit shall be provided having a hang time of not less than 1 second (s) and not more than 1.7 s so that no relay in the transmitter will be slaved to the CW key. The keyed CW wave shall be free of transients and splatter to the degree specified under spurious responses (see 3.16.8).

3.16.3.2 AM(A3h). Emission in the AM mode of operation shall be comprised of the carrier frequency and the USB portion of the modulator output. An internal control shall be provided to permit adjustment for full carrier level in the AM mode (100 percent modulation).

3.16.3.3 SSB (A3a and A3j). Emission in either A3a or A3j mode shall consist of either USB or LSB (selectable), consistent with 3.16.3.3.1.

3.16.3.3.1 Internal selection control. An internal control shall be provided to permit selection in the SSB mode for the nominal levels of carrier insertions specified in a through d:

a. Full carrier (0 dB)
b. -10 dB of carrier suppression
c. -20 dB of carrier suppression
d. Full carrier suppression

Equipment shall be delivered with the control set on full carrier suppression.

3.16.3.4 RATT (A7j). An internal RATT tone generator shall be provided to generate tone mark and space signals. The unit shall be capable of operation at speeds up to 100 words per minute when excited with a neutral input loop current of from 5 mA to 75 mA. Tones shall be shifted as specified in a and b:

a. +425 Hz, and
b. ±85 Hz, reference to a tone center frequency of 2000 Hz.

Tone shift shall be selected by a front panel switch. The output shall be as specified in c and d, respectively:

c. 2425 Hz ±20 Hz, and
d. 2085 Hz ±4 Hz, above the nominal carrier frequency with zero input loop current.

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The output frequency shall be as specified in e and f, respectively:

e. 1575 Hz ±20 Hz, and
f. 1915 Hz ±4 Hz, above the nominal carrier frequency with an input loop current of from 5 mA to 75 mA.

A connector shall be provided at the rear of the cabinet for local RATT input signal and keyline. Internal diode protection shall be provided for polarity protection. RATT transmission shall be on the USB only. The RATT generator shall deliver rated output voltage to the audio amplifier with the center tap grounded or ungrounded.

3.16.3.5 Multi-channel (A7j and A9b). Multi-channel operation shall be obtained by inserting multiple tones at the external audio input receptacles of the transmitter.

3.16.3.6 ISB (A3b). ISB suppressed carrier operation of the equipment shall be accomplished by selecting the ISB position on the front panel mode selector switch. Provision shall be included to permit front panel connection of a handset for local operation. Two remote audio input (USB and LSB) may be used in the remote mode of operation. For remote audio input operation, two connectors shall be provided on the rear of the cabinet.

3.16.3.6.1 ISB/RATT position. In addition to the ISB position on the front panel mode selector switch, an ISB/RATT position shall be provided to permit simultaneous operation of both sidebands, but with the internal RATT tone generator output on the upper sideband.

3.16.3.7 Envelope delay distortion. The maximum envelope delay distortion shall not exceed 500 microseconds (µs) throughout the region from 825 Hz to 3100 Hz, nor exceed 150 µs for any 100 Hz interval within this range. Maximum envelope delay distortion shall not exceed 700 µs in the region from 605 Hz to 3100 Hz, nor exceed 1.5 milliseconds (ms) over the complete range from 380 Hz to 3100 Hz.

3.16.4 IM distortion. All IM products shall be reduced at least 35 dB below either tone of a two-tone RF envelope at 250 mW PEP, from 2.0 MHz to 30.0 MHz, produced by using the standard two-tone input (see 6.7.3). With a two-tone output of 100 mW, all IM products shall be reduced at least 40 dB below either tone of the two-tone envelope.

3.16.4.1 Vibration test distortion. When tested in accordance with 3.15.1.23.2, any vibration related AM shall be less than 3 percent of a single-tone RF output of 62.2 mW. The RF output signal shall not break-up. Phase jitter of the unmodulated AM carrier shall not exceed the requirements of 3.16.18.1.6.3.

3.16.5 Harmonic suppression. Second harmonic output shall be reduced at least 40 dB below a single-tone output of 62.5 mW. All other harmonic outputs shall be reduced at least 45 dB below 62.5 mW.

3.16.6 Carrier suppression. In the USB and LSB modes of operation, with the carrier insertion switch in the full carrier suppression position, the carrier shall be suppressed a minimum of 50 dB below a 62.5 mW single-tone output. With the carrier reinsertion control in the 0 dB (full carrier), -10 dB, or -20 dB position, the carrier shall be within 3 dB of the selected level relative to a 62.5 mW single-tone output in the USB or LSB modes. The audio input shall be a 1300-Hz tone at 150 mW.

3.16.7 Opposite sideband suppression. The level of the opposite sideband frequencies (measured at a frequency 1300 Hz or more from the suppressed carrier) shall be suppressed a minimum of 60 dB below the level of a single-tone output of 62.5 mW.

3.16.8 Narrowband spurious RF emission. Spurious RF emissions, when measured across a standard 50-ohm resistive load at the equipment RF output terminals, shall conform to 3.16.8.1 and 3.16.8.2.

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3.16.8.1 Out-of-band outputs. All narrowband spurious emissions with the exception of harmonics of the operating frequency shall be reduced as specified in a and b, relative to a single-tone output of 10 mW to 125 mW:

a. f_o < 20 MHZ

$\frac{f_o}{1.025} < f_s < 1.10f_o$; 50 dB

$\frac{f_o}{1.10} \geq f_s \geq 1.10f_o$; 90 dB

$\frac{f_o}{1.15} \geq f_s \geq 1.15f_o$; 120 dB

b. $f_o \geq$ 20 MHz

$\frac{f_o}{1.025} < f_s < 1.10f_o$; 50 dB

$\frac{f_o}{1.10} \geq f_s \geq 1.10f_o$; 90 dB

$\frac{f_o}{1.15} \geq f_s \geq 1.15f_o$; 110 dB

Where f_o is the transmitter operating frequency, and f_s is the frequency of a spurious emission.

3.16.8.2 In-band products. Spurious emissions which fall within the information band shall be reduced at least 40 dB below a single-tone output with the levels specified in 3.16.8.1. This requirement includes frequencies which are produced by modulation of the carrier by harmonics of the tone.

3.16.9 Noise. Equipment noise emissions measured across a 50-ohm resistive load at the RF output terminals shall conform to 3.16.9.1 through 3.16.9.3.

3.16.9.1 Key-up noise. With the audio input terminated in a 600-ohm resistance, no audio input signal, and the equipment unkeyed, the in-band noise output shall not exceed -115 decibels referred to one mW (dBm) for the 3 kHz information band.

3.16.9.2 Keyed and modulated in-band noise. With a standard single-tone audio input in the mode (USB or LSB), broadband noise in the 1 Hz band within the selected 3 Hz sideband shall be a minimum of 75 dB below a single-tone RF output reference of 10 mW to 62.5 mW.

3.16.9.3 Out-of-band broad band noise. Exciter out-of-band broadband noise measured in a 3 Hz band at frequency spacings of 10 percent or greater from the operating frequency ($f_s/f_o \geq 1.10$ or $f_o/f_s \geq 1.10$) shall not exceed -120 dBm. This limit shall be applicable for any RF output at the operating frequency within the exciter rated output power range.

3.16.10 Power supply hum distortion. All hum distortion signals shall be suppressed a minimum of 45 dB with respect to an RF output of 62.5 mW produced by a standard single-tone audio input.

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3.16.11 APC. An APC circuit shall be provided to limit the average power output to a level sufficient to protect an associated RF amplifier from tube damage due to excessive plate dissipation. The APC voltage shall be derived from the associated RF amplifier.

3.16.12 Peak power control (PPC). A PPC circuit shall be provided to limit power output to a level sufficient to safeguard an associated RF amplifier and antenna coupler from damage. The PPC voltage shall be derived from the associated RF amplifier.

3.16.13 CW keying. Keying speeds up to 25 dot cycles per s (approximately 40 words per minute) shall be possible. The keyed wave shall be free from transients and splatter.

3.16.14 Keying transient. There shall be no keying transient (signal burst) when the equipment is keyed and unkeyed.

3.16.15 RATT keying. Keying speeds up to 80 baud, using total frequency shift of either:

a. 850 Hz, or
b. 170 Hz, centered on 2000 Hz, shall be possible.

External RATT keying shall be possible with inputs of:

c. 120 volts direct current (VDC) neutral, 5 mA to 75 mA (high level), or
d. Unbalanced voltage digital interface (low level) in accordance with the Unbalanced voltage digital interface circuit paragraph of MIL-STD-188-114, except that the NOTE under the Signalling sense paragraph is not applicable. Selection of high or low level input shall be an installation option and shall be accomplished by strapping. RATT emission shall be in the USB of the equipment.

3.16.16 Audio compression. An audio compression circuit shall be provided to permit operation using signals of +20 dB above standard audio input without a change in the performance specified herein. Time constants for the circuit shall be as specified in a and b:

a. Attack time (ms): Not less than (NLT) 30, not more than (NMT) 50.
b. Decay time (s): NLT 1.5, NMT 2.0

This requirement is not applicable to data mode operation.

3.16.17 Accuracy of main tuning indicator calibration. The accuracy of frequency indications shall be within the limits specified in 3.16.17.1 and 3.16.17.2. The specified accuracies shall apply to all tuning indicators, acting in combination, to indicate the tuning of the equipment.

3.16.17.1 Step tuning. Tuning in steps of 100 Hz shall be provided with an accuracy equivalent to that of the internal frequency standard reference. Tuning the exciter to any frequency within the tuning range in less than 20 s shall be possible.

3.16.17.2 Tune initiate signal. A tune initiate (ground pulse) signal, of 100 ms (minimum) duration, shall be produced by a change of any of the frequency selector knobs, except the 100 Hz selector, for tuning of an associated RF amplifier or antenna coupler.

3.16.18 Frequency and gain stability. The equipment shall include provisions for operation from either an internal or external frequency standard of 5 MHz. Frequency stability requirements apply to operations using the internal standard.

3.16.18.1 Frequency stability and gain variations. The requirement of 3.16.18.1.1 through 3.16.18.1.6.3 for frequency stability and gain variation shall apply under conditions of A1 or A3j operation, as specified. The frequency stability figures shall apply to the variation of the carrier frequency. All figures shall refer to total variations, either instantaneous, transient, or gradual changes. The requirements specified herein shall apply at any frequency within the frequency range of the equipment.

3.16.18.1.1 Sixty-minute warm-up period. Measurements shall be made in the A1 mode at constant ambient temperature, constant relative humidity and constant line input voltage. Two minutes after turn-on-time, the equipment shall be within 0.1 percent of the frequency it will have reached at the end of 60 minutes. Fifteen minutes after turn-on-time, the equipment shall be within 0.002 percent of the frequency reached 60 minutes after turn-on. Sixty minutes after turn-on, the equipment shall be within 1 part in 10^7 of the desired frequency.

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3.16.18.1.2 Variation in primary power supply voltage and frequency. Variation of primary power supply line voltage over a range of 115 V +10 percent, and of power line frequency over a range of 48 Hz to 420 Hz, in any combination, shall not cause a change in resonant frequency of the equipment by more than 2 parts in 10^8 (total) for A1 signals, or a change in overall gain in excess of 2 dB.

3.16.18.1.3 Variation in relative humidity. Exposure of the equipment to the humidity conditions specified in the Humidity paragraph of MIL-E-16400 shall not cause loss of synthesizer frequency lock or excessive changes in frequency or gain. During the checks performed at the end of the second and fifth cycle of the humidity test, the equipment shall be energized for a maximum period of 30 minutes to verify proper operation. At the end of the 30-minute period, frequency for SSB modes shall be within 2 parts in 10^7 of the nominal, and maximum permissible gain changes (APC and PPC voltages fixed) shall not produce output variations of more than 6 dB (total) from the levels recorded as reference at the end of the initial dry-out period.

3.16.18.1.4 Variations in ambient temperature. Variations in ambient temperature shall be in accordance with 3.6.18.1.4.1 and 3.6.18.1.4.2.

3.6.18.1.4.1 Nonoperating condition. The equipment shall be capable of normal operation after exposure to the temperature cycle specified in MIL-E-16400 for temperature tests (Range 4) to be applied to the equipment when in nonoperating condition.

3.16.18.1.4.2 Operating condition. The equipment, when subjected to the temperature test for Range 4 equipment specified in MIL-E-16400 (when in the operating condition), shall not exhibit frequency variations in excess of 1 part in 10^7. The maximum permissible change in gain for any mode of operation shall be 12 dB. The specified limits for variation of frequency and gain shall be the total deviation of these parameters, which occur during the time period required by the high and low operating temperature cycling test procedure specified in MIL-E-16400.

3.16.18.1.5 Control variations. Frequency variations resulting from a variation of nonfrequency determining controls throughout the full range of operation shall not exceed 1 part in 10^8. This requirement shall apply to those controls which operate in nonfrequency determining circuits and to controls which affect the RF or audio frequency gain or level. The specified limit for frequency variation shall be the absolute maximum deviation of this parameter which occurs during the manipulation of controls throughout the full range of operation.

3.16.18.1.6 Short and long term stability. The equipment shall be capable of operation for a period of 24 hours without operator attention. Without operation attention shall be interpreted as meaning that reliable transmission of a desired signal shall be assured throughout a 24-hour period without the necessity for readjustment of any tuning control. This requirement shall be applicable throughout the complete frequency range of the equipment.

3.16.18.1.6.1 Sudden changes during 24-hour period. During the 24-hour period of operation specified in 3.16.18.1.6, the equipment shall not exhibit sudden sporadic shifts in the frequency generated by any of the oscillators, or sudden changes in gain that may be the result of easing tension in tuning drive mechanisms or from similar effects.

3.16.18.1.6.2 Frequency drift rate. After an initial stabilization period, not to exceed 48 hours following turn-on, the overall frequency drift rate shall not exceed 1 part in 10^8 (total) per day. This rate shall be achieved throughout the operating frequency range of the equipment in any mode of operation. The long-term frequency drift rate, averaged over a period of 60 days, shall not exceed 2.5 parts in a 10^9 (total) per day. During measurement of frequency drift rate, changes in power supply line voltage and frequency shall be limited to +1 percent, and temperature variations shall be limited to 10° C (total). Under the same conditions, equipment gain with APC voltage constant shall not change by more than 3 dB (total) during a 24-hour period after initial stabilization.

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3.16.18.1.6.3 Phase stability. Phase stability shall be such that the probability of experiencing spurious phase changes greater than 5 degrees between successive frames of 13.3 ms or 22 ms shall be less than 1 percent. Measurement shall be the average phase in central 9.09 ms or 18.18 ms period for 13.3 ms or 22 ms frame times, respectively. Data shall be acquired for a sufficient number of adjacent frame pairs to establish the specified probability with a confidence of at least 95 percent.

3.16.19 Regeneration. There shall be no evidence of unintended oscillation or instability under any condition of equipment operation, including primary line voltage 10 percent higher than the normal primary line voltage.

3.16.20 Oscillator radiation. Oscillator radiation shall conform to MIL-STD-461, Class A4 requirements.

3.16.21 Frequency standard input, calibration and adjustment. The equipment shall contain provisions that permit the comparison of an external higher stability frequency standard of 5 MHz with the internal frequency standard of the equipment. The comparator shall function with external signal levels between 450 millivolts (mV) and 10.0 V, open circuit, in series with 50 ohms.

3.16.21.1 Frequency standard calibration. The internal frequency standard shall contain a means of indicating calibration accuracy when compared to an external frequency standard. A dial shall be provided to indicate positioning of the frequency control circuit element within the adjustment range.

3.16.21.1.1 Frequency standard adjustment range. The internal 5 MHz frequency standard shall be capable of adjustment over a minimum frequency range of 5 MHz ±20 Hz. The adjustment shall be accessible from the top surface of the frequency standard assembly without removing the assembly cover. Resolution of the adjustment shall be adequate to permit setting to within 1 part in 10^9. Piston type adjustments shall be protected from damage at the ends of the adjustment ranges.

3.16.21.2 Operation using an external frequency standard. The equipment shall be capable of operation from an external frequency standard of 5 MHz with any input level between 500 mV and 10.0 V, open circuit in series with 50 ohms. When the equipment is operated from an external frequency standard and the level of the externally supplied 5 MHz signal falls below 500 mV ±100 mV open circuit, in series with 50 ohms, the internally contained 5 MHz crystal oscillator shall be automatically selected as the reference (see 3.16.21.4). When the externally supplied 5 MHz signal is restored to a level greater than 800 mV ±100 mV, the external 5 MHz signal shall again be automatically chosen as the reference. Input impedence at the external 5 MHz input terminals shall be 50 ohms, nominal within a voltage standing wave radio (VSWR) of 1.5:1 or less.

3.16.21.3 Reference output signal. The equipment shall provide a 5 MHz reference output signal of between +7 dBm and +13 dBm, delivered to a 50-ohm load, for external use.

3.16.21.4 Frequency standard switch functions. The internal frequency standard assembly shall have a function switch located on the top of the assembly, accessible without removal of the assembly dust cover. The switch shall have three positions with the functions (clockwise) specified in a through c:

a. EXT (OVEN STBY) (External Oven standby): Provides operation from an external frequency distribution system; the internal frequency standard oscillator and oven are kept on for immediate availability with full accuracy.

b. EXT NORM (External Normal): Provides operation from an external frequency distribution system; the internal frequency standard oscillator and oven are kept on for immediate availability with full accuracy.

c. INT OR COMP (Internal or Compare): Used when external frequency distribution system is not available or not functioning, or for testing.

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3.16.22 Frequency generation. The frequency generation process in the translator and synthesizer assembly shall utilize only a single reference frequency, with all internally-generated frequencies used in the synthesis of the operating frequency derived by arithmetic operations on, and phase locked to the reference. The synthesizer shall not use multi-crystal generators or motor driven frequency selectors.

3.16.23 Overall bandpass characteristics. Measured from audio input to RF output, the overall bandpass characteristic shall comply with the requirements specified in a through d:

a. Response shall be down 3 dB or less below the peak response from carrier +300 Hz to carrier +3500 Hz for USB or from carrier -300 Hz to carrier -3500 Hz for LSB.
b. Response shall be down 20 dB or more below the peak response from carrier +50 Hz and below and carrier +4000 Hz and above for USB, and at carrier -50 Hz and above and carrier -4000 Hz and below LSB.
c. Response shall be down 60 dB or more below the peak response at carrier +5300 Hz or above for USB, and at carrier -5300 Hz or below for LSB.
d. Maximum peak-to-valley undulations over the 300 Hz to bandpass shall be 3 dB.

3.16.24 Tune cycle. The voltage from the regulated +20 VDC power supply shall drop to 0 VDC and remain there from the start of resetting the 1 MHz and 10 MHz frequency controls to completion of tuning of the RF amplifier turret and MHZ synthesizer drive motor, when the 1 MHz and 10 MHz frequency controls are tuned at a rate no slower than two steps per second.

3.16.25 Data keying. The exciter shall be continuously keyed when in the data mode.

3.16.26 Audio input for data. Equipment data audio input shall be in accordance with TADIL A of MIL-STD-203-1. Input level shall be 0 dBm average in each sideband, with a maximum peak to average ratio of 11.6 dB. The equipment audio input impedance for each sideband shall be 600 ohms, +10 percent throughout the range from 605 Hz to 2915 Hz. Input transformer primary windings shall be centertapped, ungrounded, and have a maximum FDC resistance of 30 ohms each side of the center tap. The transmitter data audio input circuits shall be equipped with screwdriver adjustable gain control to permit proper data operation over an input amplitude range of 0 dB +5 dBm average in each sideband. The controls shall be readily accessible for adjustment during installation and maintenance.

3.16.26.1 Audio and keyline connections for data. Provisions shall be made to accept data audio and keyline signals from either the data modem connector (see 3.4.c.4) or the transmitter switchboard connector on an associated AM-3924( )/URT Power Amplifier. Connection of the desired source of data audio and keyline signals to the data audio processing and data keyline detector circuits shall be accomplished as an installation option. For this purpose, a readily available terminal board shall be provided to permit selection of the desired source by suitable strapping.

3.16.27 Electromagnetic compatibility (EMC). The equipment shall conform to the requirements of MIL-STD-461 for Class A4, C-E equipment.

3.16.28 Accelerated life. The equipment shall conform to the Accelerated life paragraph of MIL-E-16400.

3.17 Quantitative maintainability requirements. This requirement applies to the GFE design. If the GFE design does not conform to the requirement, redesign of the equipment shall preserve compatibility with the GFE. The equipment shall have a mean-corrective-maintenance-time not exceeding 1.5 hours and a maximum-corrective-maintenance-time (M_{maxct}) not exceeding 4.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, PCBs) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The mean-time-to-repair (MTTR) includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The equipment is in a corrective maintenance mode whenever a failure, including a malfunction that allows degraded operation, occurs.

3.18 Workmanship. Workmanship shall be as specified in 3.18.1 and 3.18.2.

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3.18.1 General workmanship. The workmanship and soldering requirements shall be in accordance with the Workmanship and Soldering paragraphs of MIL-E-16400.

3.18.2 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration (see 4.5.2.2.1) at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.5.2.2.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all irspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 General inspection. Inspection shall be as specified in MIL-E-16400, except as otherwise specified in 4.2.1 through 4.4.4.

4.2.1 Classification of inspections. The inspection requirements specified herein are classified as specified in a through e:

a. First article inspection (see 4.3)
b. Quality conformance inspection:
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

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TABLE II. Examination and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination	3.1.2	4.5.1	X	X		
Finish	3.1.2	4.5.1	X	X		
Parts and materials	3.1.2	4.5.1	X	X		
Steady-state voltage	3.6	4.5.6, 4.5.7.21	X		X	
Power, power factor	3.6	4.5.1, 4.5.7.1	X		X	
Steady-state frequency	3.6.1	4.5.6, 4.5.7.21	X		X	
Transient voltage	3.6.1	4.5.6, 4.5.7.21	X		X	
Power interruption	3.6.1	4.5.1, 4.5.7.22	X		X	
Electrical overload	3.6.2	4.5.5	X		X	
Leakage current	3.6.3	4.5.3	X			
Radio set system	3.9	4.5.5	X			
MUTE interface	3.9.2	4.5.7.24	X	X		
Weight	3.10	4.5.1	X		X	
Dimensions	3.10	4.5.1	X		X	
Temperature sequence	3.11	4.5.1, 4.5.7.18	X			X
Low temperature	3.11	4.5.1, 4.5.7.18	X			X
High temperature	3.11	4.5.1, 4.5.7.18	X			X
Humidity	3.11	4.5.1, 4.5.7.18	X			X
Salt fog	3.11	4.5.1	X			X
Reliability	3.13	4.6				
Mechanical design and construction	3.15	4.5.1	X		X	
Drawer orientation	3.15.1.1	4.5.5	X	X		
Enclosure	3.15.1.3	4.5.1, 4.5.7.20	X			X
Bonding and grounding	3.15.1.4	4.5.1	X	X		
Corrosion protection	3.15.1.5	4.5.1	X		X	
Painting	3.15.1.5.1	4.5.1	X		X	
Mode selector switch	3.15.1.8.1	4.5.5	X		X	
Assembly identification	3.15.1.18	4.5.5	X		X	
Shock	3.15.1.23.1	4.5.6, 4.5.7.19	X			X
Vibration	3.15.1.23.2	4.5.6, 4.5.7.19	X			X
Inclination	3.15.1.23.3	4.5.6, 4.5.7.19	X			X
Welding	3.15.1.25	4.5.1	X		X	
Conductors	3.15.1.28	4.5.1	X		X	
Identification and marking	3.15.1.29	4.5.1	X	X		
Thermal design	3.15.1.34	4.5.1	X			
Frequency standard construction	3.15.1.35	4.5.5	X		X	
Maintainability	3.17	4.7				X
Power output	3.16.1	4.5.6, 4.5.7.1	X	X	X	
Maximum RF output	3.16.1.1	4.5.6, 4.5.7.1	X		X	

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TABLE II. Examination and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Power output leveling	3.16.2	4.5.6, 4.5.7.2	X	X	X	
Sideband equalization	3.16.2.1	4.5.6, 4.5.7.2.1	X	X		
CW hold	3.16.3.1	4.5.6	X		X	
Carrier insertion	3.16.3.3.1	4.5.6	X		X	
RATT tone shift	3.16.3.4	4.5.6	X		X	
Envelope delay distortion	3.16.3.7	4.5.6, 4.5.7.16	X		X	
IM distortion	3.16.4	4.5.6, 4.5.7.3	X	X	X	
Harmonic suppression	3.16.5	4.5.6, 4.5.7.4	X	X	X	
Carrier suppression	3.16.6	4.5.6, 4.5.7.4	X	X	X	
Opposite sideband suppression	3.16.7	4.5.6, 4.5.7.6	X	X	X	
Out-of band spurious	3.16.8.1	4.5.6, 4.5.7.7	X		X	
In-band spurious	3.16.8.2	4.5.6, 4.5.7.7	X		X	
Noise	3.16.9	4.5.6, 4.5.7.8	X		X	
Power supply hum distortion	3.16.10	4.5.6, 4.5.7.9	X	X	X	
APC	3.16.11	4.5.5	X		X	
PPC	3.16.12	4.5.5	X		X	
CW keying	3.16.13	4.5.6	X		X	
Keying transient	3.16.14	4.5.5	X		X	
RATT keying	3.16.15	4.5.6	X		X	
Audio compression	3.16.16	4.5.6, 4.5.7.10	X		X	
Accuracy of tuning	3.16.17	4.5.6, 4.5.7.11	X		X	
Tuning time	3.16.17.1	4.5.6, 4.5.7.11.1	X		X	
Tune initiate pulse	3.16.17.2	4.5.6	X		X	
Sixty-minute warm-up	3.16.18.1.1	4.5.6, 4.5.7.12.1	X		X	
Line voltage variation	3.16.18.1.2	4.5.1, 4.5.7.12.2	X		X	X
Line frequency variation	3.16.18.1.2	4.5.1, 4.5.7.12.2	X		X	X
Variation in humidity	3.16.18.1.3	4.5.1, 4.5.7.18	X			X
Variation in temperature	3.16.18.1.4	4.5.1, 4.5.7.18	X			X
Control variation	3.16.18.1.5	4.5.6	X		X	
Frequency drift rate	3.16.18.1.6.2	4.5.6, 4.5.7.12.3	X		X	
Phase stability	3.16.18.1.6.3	4.5.6, 4.5.7.13	X		X	
Regeneration	3.16.19	4.5.5	X		X	
Oscillator radiation	3.16.20	4.5.6, 4.5.7.14	X		X	

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TABLE II. Examination and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Frequency standard calibration	3.16.21.1	4.5.6	X		X	
Adjustment range	3.16.21.1.1	4.5.6	X		X	
External frequency range	3.16.21.2	4.5.6	X		X	
Overall bandpass characteristics	3.16.23	4.5.6	X		X	
Tune cycle	3.16.24	4.5.6	X		X	
Data keying	3.16.25	4.5.5	X		X	
Audio input for data	3.16.26	4.5.5	X		X	
EMI	3.16.27	4.5.6, 4.5.7.15	X		X	
Accelerated life	3.16.28	4.5.1, 4.5.7.23	X			
Workmanship	3.18.1	4.5.2.1	X	X		
Soldering	3.18.1	4.5.2.1	X		X	
Random vibration	3.18.2	4.5.2.2.1	X	X		
Temperature cycling	3.18.2	4.5.2.2.2	X	X		

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A. Production inspection shall be conducted on each equipment offered for delivery. Production inspection shall encompass such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls or adjustments and tests which detect hidden defects of materials. Production inspection shall include the examination and tests shown in Group A or TABLE II, subject to the conditions of 4.4.1.1.

4.4.1.1 Group A tests. The Group A tests shall be performed in accordance with the test methods of 4.5.7, except as modified as specified in a through f:

a. Power output and IM distortion measurements shall be made at the center frequency of each primary band in the USB mode. Measurements shall be made in the remaining modes at the frequency exhibiting the lowest power output and worst IM distortion
b. Power leveling and signal-to-hum ratio: 2 MHz and 29 MHz bands
c. Harmonic suppression: second and third order, 2 MHz and 29 MHz bands
d. Carrier and opposite sideband suppression: 2.1 MHz, USB and LSB modes
e. Spurious emissions: explore three MHz bands of highest level spurious output determined by first article inspection
f. Frequency stability: 5 MHz

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4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in conformance with the inspection procedures of MIL-STD-105, using inspection level S-3 with an acceptable quality level of 6.5 percent for each attribute. Production control inspection shall consist of examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; test to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and tests shown in Group B of TABLE II. Production control inspection shall be conducted on inspection lots that have passed the production inspection of 4.4.1.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units and equipment as a whole; life tests, simulated service tests; tests on the effect of changes of environment (such as extremes of temperature and humidity, effect of salt air); and test of the effects of shock, vibration and inclination. Environmental inspection shall include the examinations and tests shown in Group C of TABLE II.

4.4.3.1 Sampling for inspection of equipment. The equipment selected for environmental inspection shall pass all tests, except EMI, prior to acceptance of a lot from which the sample is selected. One equipment from the first month's production list, or from the first 50 units produced, shall be subjected to environmental inspection. One equipment from each successive production lot of 100, or one per month if less than 100 are produced monthly, shall be subjected to environmental inspection. One equipment from each 200 equipments produced shall be subjected to EMI test. Environmental inspection shall be performed on equipments that have passed the production inspection of 4.4.1.

4.4.3.2 Nonconforming environmental inspection sample unit. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the correction action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of the failure, the matter shall be referred to the contracting officer (see 6.6).

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to and passed production control or environmental inspection, or both, may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all defects.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.7.24.

4.5.1 Examinations and tests. When 4.5.1 is specified in TABLE II, the test method specified in MIL-E-16400 shall be applied.

4.5.2 Workmanship. Workmanship shall be as specified in 4.5.2.1 through 4.5.2.2.2.

4.5.2.1 General workmanship and soldering. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.18.1. Each solder connection and associated wiring or leads shall be visually examined.

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4.5.2.2 Workmanship screen. Prior to conducting temperature cycling, vabration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.5.2.2.1 Random vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 1: Input vibration levels shall be measured at the mounting point of the items under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth in accordance with MIL-STD-810C, Method 514.2, Random vibration test paragraph, shall be used for control and analysis of the acceleration spectral density. The instantaneous acceleration peaks shall be limited to three times the acceleration rms value. All failures occurring during screening shall be corrected and the vibration resumed.

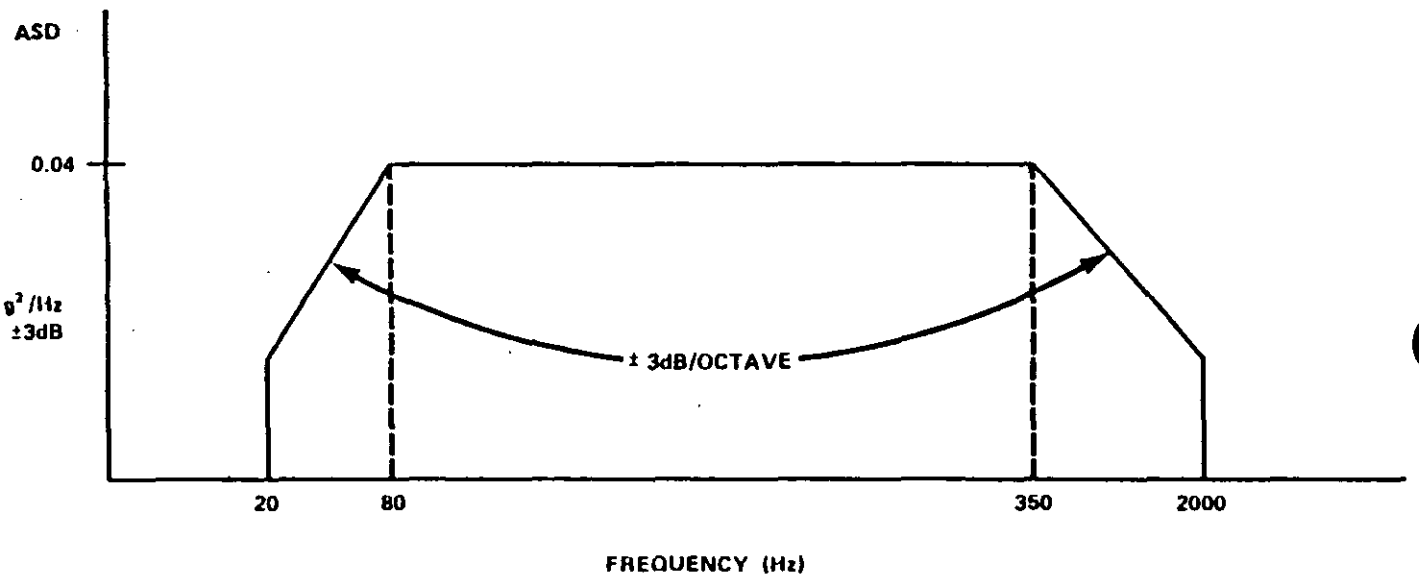


FIGURE 1. Random vibration curve.

4.5.2.2.2 Temperature cycling. Each equipment shall be subjected to 10 cycles of the temperature curve indicated in FIGURE 2. The temperature rate of change shall be not less than 5°C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electical or electronic part in the unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

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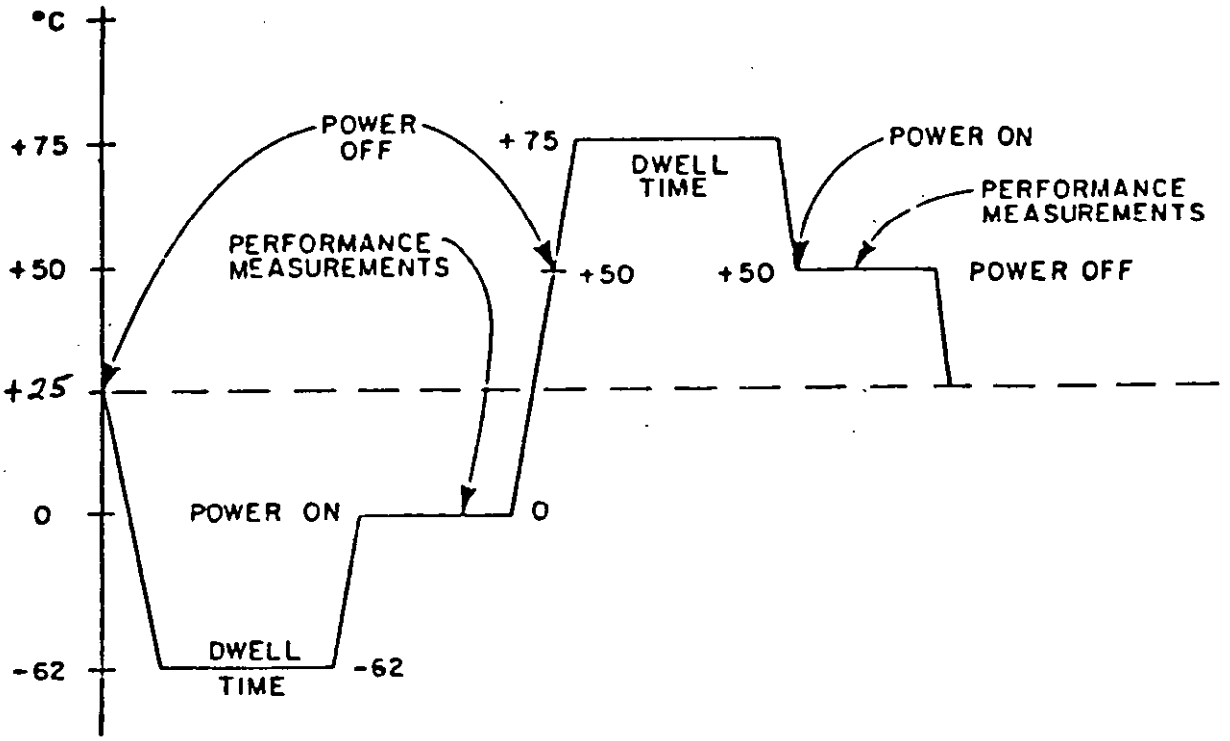


FIGURE 2. Temperature curve (one cycle) (not to scale).

4.5.3 Leakage current test. The leakage current test shall be as specified in 4.5.3.1 and 4.5.3.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNDERGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBLITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.3.1 Equipment connections. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 3 if connected to single-phase source power, and as shown in FIGURE 4 if connected to three-phase source power.

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4.5.3.2 Measurement. Leakage current shall be measured on the equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all controls, and control shafts with knobs removed and the voltage measured for every combination of switch positions available in FIGURE 3 and FIGURE 4. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.4 Cables. Interconnecting cables to be supplied with the end items shall be installed on the end item prior to any testing, and shall be used for the duration of all testing.

4.5.5 Contractor validation. The contractor shall inspect the equipment to determine technical compliance and adequacy for the specified characteristics or requirements. This validation shall include qualitative examination and testing whenever a function is specified.

4.5.6 Inspection by testing. When this paragraph is specified in TABLE II, the contractor shall perform quantitative testing.

4.5.7 Detailed test methods. Detailed test methods shall be as specified in 4.5.7.1 through 4.5.7.24.

4.5.7.1 Power output tests. Power output measurements shall be made at the low gain frequency (see 4.5.7.2) of each MHz band in USB, LSB, ISB, CW, RATT and AM modes. A standard single tone audio input to each of the sidebands shall be used for ISB measurements. RATT measurements may be made with zero loop current. The unmodulated carrier shall be measured in the AM mode.

4.5.7.2 Power output leveling tests. Key the equipment in the CW mode. Adjust DC bias voltage applied to the APC line to hold the RF output power into a 50-ohm load below voltage limiting level (250 mW). For each MHz band record the setting of the 100 kHz, 10 kHz, 1 kHz, and 100 Hz frequency controls and that produce the highest and lowest RF power. After the equipment has reached a stable operating temperature (on for at least 4 hours), measure RF output power at all frequencies recorded in the previous step with APC voltage held constant within 0.0001 VDC. Calcuate the dB power difference between the high and low gain frequency in each MHz band and the difference between the overall high and low gain frequency.

4.5.7.2.1 Sideband output equalization tests. Equalization of overall transmit gain in each sideband shall be measured with APC adjusted to prevent limiting. A single tone shall be placed in each sideband at the peak of the passband ripple.

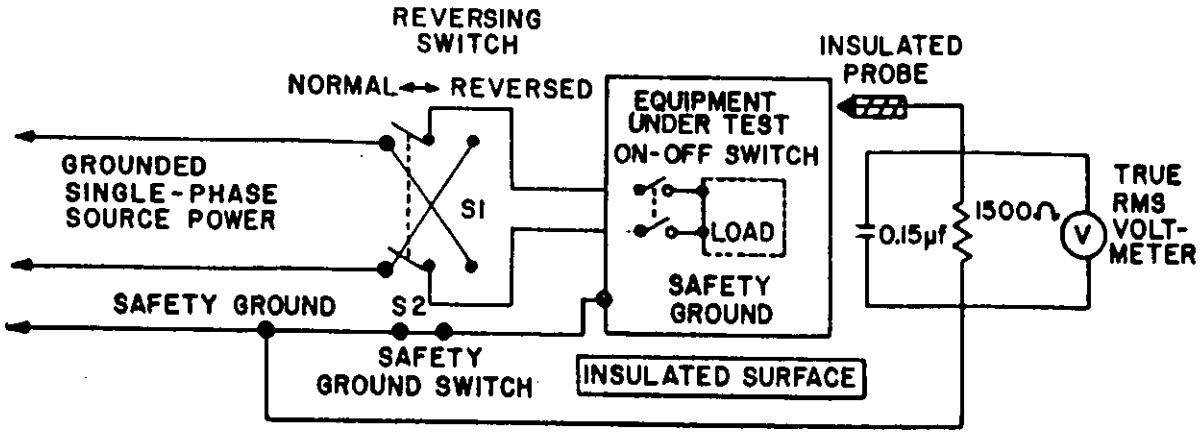
4.5.7.3 IM distortion tests. IM distortion shall be measured by applying a standard two-tone audio input (see 6.7.3) and monitoring the output developed across a standard dummy load (50 ohm) with a spectrum analyzer. Measurements shall be made in the A3j mode using the standard two-tone audio input specified in 6.7.3. Measurements shall be made at operating frequencies of 2 MHz, 29.9 Mhz and at the high and low gain frequencies. For each test frequency, measurements shall be made with APC voltage adjusted to produce outputs of 100 mW and 250 mW PEP.

4.5.7.4 Harmonic suppression test. Harmonic outputs shall be measured for operating frequencies of 2 MHz, 4 MHz, 8 MHz, 16 MHz, and 29.9 MHz.

4.5.7.5 Carrier suppression test. Measurements shall be made at 2.1 MHz and 29.9 MHz.

4.5.7.6 Opposite sideband suppression tests. Measurements shall be made at 2.1 MHz, 17.1 MHz, and 29.9 MHz.

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GENERAL ORDER OF TEST:

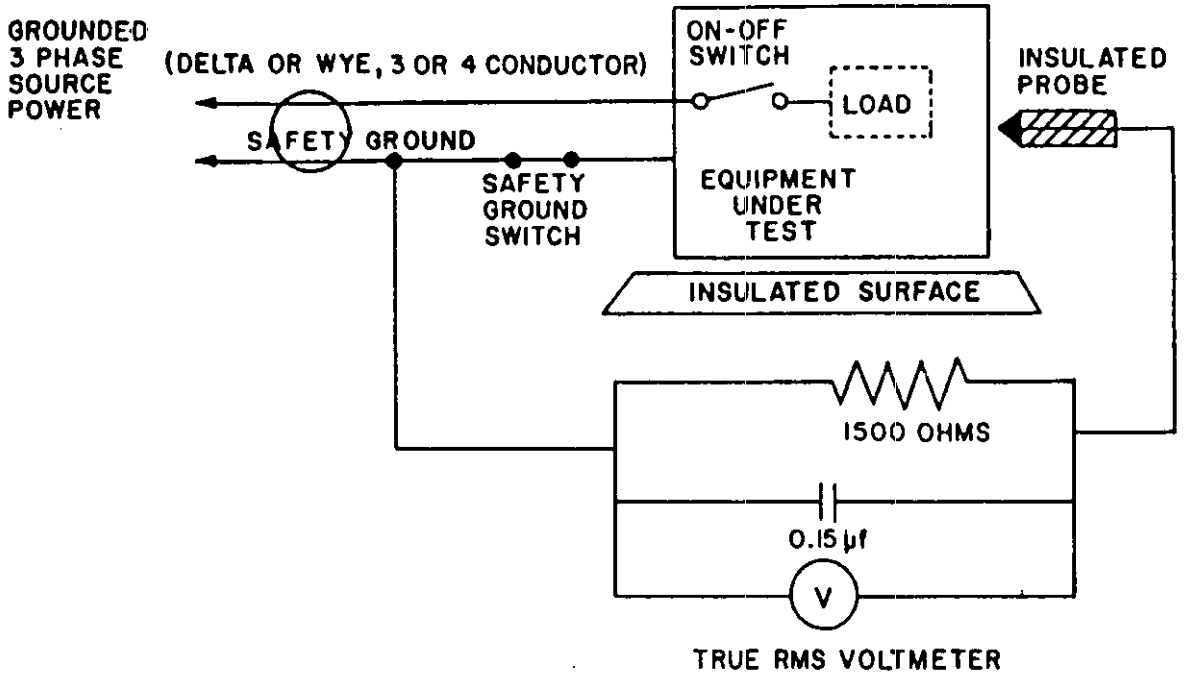
1. Source power OFF. Connect equipment per diagram.
2. ON-OFF SW OFF. S1 SW normal. S2 SW closed. Connect source power.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. For each probe point, record voltmeter reading (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. Repeat Step 4.
6. S1 SW REVERSED. ON-OFF SW ON. Repeat Step 4.
7. ON-OFF SW OFF. Repeat Step 4.
8. S2 SW CLOSED. S1 SW normal.
9. Repeat Step 3 THRU 8 for each mode of operation.
10. Remove source power. Disconnect equipment. Sign record sheet.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 3. Single-phase test diagram for leakage current measurement.

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GENERAL ORDER OF TESTS:

1. Source power OFF. Connect equipment per diagram.
2. ON-OFF SW OFF. Safety ground switch closed. Connect source power.
3. OBSERVE WARNING Safety ground switch OPEN. ON-OFF SW ON.
4. For each probe point, record voltmeter reading (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. Repeat Step 4.
6. Safety ground switch closed.
7. Repeat Step 3 THRU 6 for each mode of operation.
8. Remove source power. Disconnect equipment. Sign record sheet.

NOTE: 1. All three phases shall be connected during measurement.
2. The safety ground conductor shall not carry load current.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 4. Three-phase test diagram for leakage current measurement.

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4.5.7.7 Spurious RF emission test. A search for spurious products shall be made throughout the complete 2 MHz to 30 MHz operating frequency range in increments of not greater than 100 kHz. For each operating frequency, a search for coherent spurious products shall be made from 0 Hz to $2f_0$ with a 10-mW single-tone RF output produced by a standard single-tone audio input. For each spurious product detected which is less than 70 dB below the reference, behavior of the spurious product shall be investigated as a function of each smaller operating frequency increment to determine worst-case performance for the spurious product. For spurious products which change in amplitutude with change of the APC voltage or audio input voltage, additional measurements shall be made to verify compliance with 3.1.6.8.

4.5.7.8 Key-up noise. Key-up noises shall be measured for operating frequencies of 2 MHz, 17 µHz and 24 MHz.

4.5.7.8.1 Keyed and modulated in-band noise. Keyed and modulated in-band noise shall be measured at the equipment high gain frequency, in upper and lower sidebands. Measurements shall be made within the region from f_0 +500 Hz to f_0 +3 kHz for USB, and f_0 -500 Hz to f_0 -3 kHz for LSB.

4.5.7.8.2 Out-of-band broadband noise. Exciter out-of-band broadband noise shall be measured in the USB mode for operating frequencies of 2.1 MHz, 17.1 MHz and 29.9 MHz. Measurements shall be made with a standard single-tone audio input and with the APC voltage adjusted to produce an RF output of 62.5 mW. Measurements shall be made on both sides of each operating test frequency.

4.5.7.9 Output signal-to-hum ratio tests. Output signal-to-hum ratio shall be measured at 2.1 MHz, 17.1 MHz, and 29.9 MHz on both USB and LSB. Measurements shall be made on both sides of each operating test frequency.

4.5.7.10 Audio characteristics test. The audio circuits shall be tested for frequency and gain characteristics. Speech processing shall be tested to ensure that essentially constant AM, allowing for changes due to speech factors, is maintained for variation (within the limits of microphone input voltages) in speech intensity and such variation shall not cause variations in power output.

4.5.7.10.1 Data audio and keyline characteristics. Measurements shall be made to verify compliance with requirements of 3.16.26 and 3.16.26.1. Audio input impedance shall be measured over the specified frequency range, and ability of the data input circuit gain adjustments to provide proper data operation over a range of input amplitudes from -5 dBm to +5 dBm shall be verified for each sideband.

4.5.7.11 Test for accuracy of main tuning indication calibration. Measurement shall be made in the A1 mode at frequencies required to verify proper operation of each frequency selector in each digital position.

4.5.7.11.1 Tuning time test. Measurement shall be made of the time required to switch from 2.0000 MHz to 29.9999 MHz.

4.5.7.12 Frequency and gain stability tests. Frequency and gain stability shall be measured in accordance with 4.5.7.12.1 through 4.5.7.12.4.

4.5.7.12.1 Sixty-minute warm-up test. The warm-up period shall begin after exposure of the equipment to the ambient temperature for a period of not less than 6 hours with no power applied, to establish the temperature at the same value as the ambient. Frequency measurements shall be made at intervals of not more than 2 minutes for the first 12 minutes, with the first measurement made two minutes after power is applied. Measurements shall be made at intervals of not more than 5 minutes for the next 18 minutes, and at intervals of not more than 10 minutes for the last 30 minutes.

4.5.7.12.2 Line voltage and frequency variation test. Measurements shall be made in accordance with the procedure specified in MIL-E-16400. A minimum stabilization period of 30 minutes shall be observed following each change of supply voltage and frequency.

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4.5.7.12.3 24-hour frequency drift test. All equipment power shall be removed for a period of at least 6 hours prior to stability measurements. The equipment shall be energized for a stabilization period not to exceed 48 hours, and operated during the test period with temperature, humidity, and line voltage controlled as specified in 3.16.18.1.6.2.

4.5.7.12.4 24-hour gain stability test. Gain stability shall be measured in A3j modes. Equipment conditioning periods, and temperature, humidity and line voltage control, shall be as specified in 4.5.7.12.3.

4.5.7.13 Phase stability test. Overall phase stability of the exciter shall be measured to verify the requirements of 3.16.18.1.6.3. Measurements shall be made with an averaging tone of 9.09 ms, with sufficent number of measurement pairs to establish the specified probability of occurence of phase changes greater than 5 degrees with a confidence of at least 95 percent.

4.5.7.14 Oscillator radiation test. Measurements shall be made at the nominal end frequencies of each primary (MHz) band.

4.5.7.15 EMC. Compliance with 3.16.27 shall be verified by test performed in accordance with MIL-STD-462.

4.5.7.16 Envelope delay distortion tests. Measurements shall be made in USB and LSB modes at any suitable radio operating frequency. An envelope delay test set shall be used to generate an audio carrier which shall be swept from 300 Hz to 310 Hz. The audio carrier shall be 50 percent amplitude modulated at 83.3 Hz, and the resulting signal shall be applied (sequentially) to the USB and LSB audio terminals. The corresponding USB and LSB RF output shall be sequentially translated to baseband and returned to the envelope delay for each sideband. An XY plotter shall be used to provide a continuous record of envelope delay as a function of frequency for each sideband.

4.5.7.17 Overall bandpass. Measurements shall be made in the ISB mode at an operating frequency of 17.5500 MHz. APC voltage shall be held constant at a value which prevents RF output limiting, and the level of the applied constant amplitude single-tone audio modulating signal shall be below the threshold of audio processing circuits. Measurements shall be made using a tracking loop arrangement to provide a continuous X-Y plot of each sideband. Measurement system accuracy, resolution and dynamic range shall be adequate to verify bandpass, ripple and outboard stop band rejection (-60 dB point) requirement of 3.16.23.

4.5.7.18 Temperature and humidity tests. Tests shall be conducted in accordance with MIL-E-16400. Measurements of gain and frequency stability shall be made to determine compliance with the requirements of 3.16.18. Measurements shall be made in USB, LSB and CW modes for a minimum of one high band and one low band frequency. Frequency accuracy (synthesizer lock) checks, phase jitter and envelope delay measurements shall be made during temperature operating tests at the end of the period of operation at 0°C after 4 hours of operation with the exciter stabilized at 50°C, and after return to room temperature. Phase jitter measurements during temperature operating tests shall be limited to operating frequencies of 2.000 MHz, 17.0000 MHz and 29.9999 MHz. During the last hour of high temperature operating test, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the equipment under test shall be controlled by 50°C by radiation-baffled heating and cooling sources as required. This is intended to simulate the worst-case shipboard thermal environment (see 4.5.7.18.1). During humidity tests, measurements shall be made at the end of the second and fifth cycles (see 3.16.17 and 4.5.7.11) .

4.5.7.18.1 Chamber temperature control provisions. The sources for maintaining the chamber environment shall be arranged or suitably baffled so that no surface at a temperature other than the desired ambient is visible to the unit under test. The maximum allowable temperature difference between any two points in the chamber shall be 10°C. The chamber control provisions shall be capable of maintaining the temperature indicated by a selected chamber ambient sensor constant with ±1°C of any selected test temperature.

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4.5.7.19 Shock, vibration and inclination tests. The shock, vibration and inclination tests shall be performed to verify compliance with 3.15.12.3.1 through 3.15.12.3.3. Prior to and after completion of shock, vibration and inclination tests, the tests specified in a through c shall be performed.

a. Power output and leveling measurements in accordance with 4.5.7.1 and 4.5.7.2 and the A1 modes for the 2 MHz, 17 MHz, and 29 MHz bands only.
b. Synthesizer locks checks in accordance with 4.5.7.11.
c. IM distortion in accordance with 4.5.7.3. During vibration tests, equipment operation and monitoring shall be in accordance with requirements of 3.16.4.1. Continuous key-down operation is required during all tests, unless otherwise specified herein.

4.5.7.20 Enclosure tests. Gain and frequency stability shall be measured prior to and after testing to determine adherence to limits specified in 3.16.18.1.6.

4.5.7.21 Supply line voltage and frequency variation and transients test. Reference power output, gain and frequency shall be measured prior to start of testing. Variation of transmitter frequency and gain with line voltage and frequency shall be measured against the requirements of 3.16.18 for nominal line voltage of 115 VAC, and for nominal line frequencies of 60 Hz and 400 Hz. Following performance of tests specified in MIL-E-16400, permanent changes shall not be made in power output, frequency and gain.

4.5.7.22 Power interruption and spike test. Power interruption tests shall be conducted in accordance with the Power interruption test paragraph of MIL-E-16400. The 2500 V spike test shall be performed in accordance with the requirements of DoD-STD-1399, Section 300. Source impedance for the spike shall be 25-ohms, resistive. Spikes shall be delivered line-line and line-ground for each primary power line. For each test, a positive-going spike shall be applied at the positive crest of the supply line voltage waveform and at the zero axis crossing and a negative-going spike shall be applied at the negative crest.

4.5.7.23 Accelerated life test. Testing shall be in accordance with the Accelerated life paragraph of MIL-E-16400. The measurements specified in a and b shall be performed at least once per day.

a. Power output and intermodulation distortion at 250 mW PEP at operating frequencies of 2 MHz, 17 MHz and 29 MHz.
b. Accuracy of main tuning indicator (4.5.8.11).

4.5.7.24 MUTE interface. The T-287( )/URT shall be tested for compliance with the MUTE interface requirement of 3.9.2.

4.6 Reliability qualification tests. A reliability qualification test shall be performed in accordance with Test Plan IVC of MIL-STD-781 on at least two units from the first months production. Environmental conditions and operational duty cycles for reliability qualification testing shall be as specified in 4.6.1 through 4.6.5.2.

4.6.1 Failure. After the initial start of the test, any parameter deviation which requires adjustment or alignment of the equipment under test shall be considered as a failure (see 6.7.7).

4.6.2 Environmental conditions for reliability testing. The environmental conditions specified in a through c shall be maintained throughout the reliability qualification test:

a. Ambient temperature shall be +50°C, +0°C, -3°C
b. Vibration shall be 1g ±10 percent peak acceleration value at least 10 minutes during each hour of equipment operating time
c. Input voltage shall be 115 VAC, 60 Hz or 400 Hz. Input voltage shall be cycled in accordance with the three-day cycle specified in 1 through 3:
1. Nominal
2. Nominal +10 percent
3. Nominal -10 percent

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4.6.3 Performance tests. Performance measurements shall be made at least daily to verify the performance characteristics specified in a through f:

a. CW (A1) mode
   1. Keying speed shall be at least 32 baud
   2. Output shall be at least 125 mW average power into 50-ohm load in key-down condition
   3. Keyed wave shall be free from transients and splatter

b. AM (A3h) mode
   1. The internal carrier reinsert control shall not be adjusted during the test providing carrier power level remains within 62.5 mW ±1.5 dB in a 50-ohm load. If carrier power variations exceed the specified limits, sufficient adjustment range of the carrier reinsert control to adjust the carrier power to 62.5 mW shall be demonstrated
   2. It shall be possible to apply an audio tone of 1000 Hz to USB input and adjust its level so that the sideband equals the carrier (62.5 mW). Without changing any controls on the equipment, sideband power shall not vary by more than 3 dB when the sideband frequency is changed from 300 Hz to 3500 Hz and the input level is maintained constant at the level required for each sideband and carrier output at 1000 Hz input
   3. Under the test conditions of b.2, the distortion products shall be at least 35 dB below the carrier

c. SSB (A3j) mode
   1. With a standard two-tone signal applied to the upper sideband remote input, the output of the transmitter shall be 250 mW PEP into a standard dummy load
   2. With modulation frequencies of 1000 Hz and 1625 Hz, the intermodulation products shall not be less than 35 dB below each reference tone for an output of 100 mW PEP into a standard dummy load
   3. Opposite sideband suppression shall be at least 60 dB, when a 150 mV, 1300 Hz tone is inserted into the remote input and the output power is 62.5 mW into a standard dummy load
   4. Carrier suppression shall be adjustable to at least 50 dB below a sideband tone of 1300 Hz having an input level of 150 mV applied to the upper sideband remote input
   5. Repeat 1 through 4 with the inputs applied to the lower sideband remote control

d. ISB (A3b) mode
   1. With a standard two-tone signal applied to each remote audio sideband input, the output of the transmitter shall be 62.5 mW PEP per sideband into a standard dummy load
   2. With modulation frequencies of 1000 Hz and 1625 Hz, the intermodulation products shall be not less than 35 dB below each reference tone for an output of 100 mW PEP into a standard dummy load
   3. Carrier suppression shall be adjustable to at least 50 dB below a sideband tone of 1300 Hz having an input level of 150 mV applied to each of the remote audio inputs

e. RATT (A7j) mode
   1. Keying rate shall be at least 80 baud
   2. Frequency shall be +425 Hz, +20 Hz, and 85 Hz ±4 Hz shift from 2000 Hz with input neutral keying of 120 VDC 60 mA nominal

f. Frequency stability. The drift of the reference oscillator (internal frequency standard) shall not exceed 1 part in 10^7 per day nor 1 part 10^6 per month

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4.6.4 Test frequencies. During the reliability tests, at least one frequency shall be tested in each band as specified in a through d:

a. F1 = 2.00015 MHz
b. F2 = 5.3340 Mhz
c. F3 = 10.6655 MHz
d. F4 = 29.9990 MHz

4.6.5 Test schedule. The test schedule of 4.6.4, including the input line voltage schedule of 4.6.2 shall be conducted. The test shall consist of three 8-hour schedules per day, the first being manned during which performance measurements shall be made. The second two may be unmanned and pertinent data shall be automatically recorded only to the extent that instrumentation permits.

TABLE III. Test schedule.

Manned schedule	Unmanned schedule
0 hours to 2 hours equipment ON 2 hours to 3 hours Performance Measurements 3 hours to 4 hours equipment OFF 4 hours to 6 hours equipment ON 6 hours to 7 hours equipment measurements 7 hours to 8 hours equipment OFF	0 hours to 3 hours equipment ON 3 hours to 4 hours equipment OFF 4 hours to 7 hours equipment ON 7 hours to 8 hours equipment OFF

4.6.5.1 Manned transmitting schedule. The mode of operation and frequency of transmission for each of the transmitting ON periods of the manned schedules shall occur in the sequence in TABLE IV.

TABLE IV. Transmitting schedules.

Day 1/	Frequency	Mode of transmission
1	F1 first 4 hours of manned schedule	CW (A1) 100 mW average power keyed at a rate of 32 baud
	F2 remainder of the day	
2	F3 first 4 hours of manned schedule	AM (A3h) 62.5 mW carrier modulated 30 percent to 50 percent at 1000 Hz
	F4 remainder of the day	
3	F1 first 4 hours of manned schedule	ISB (A3b) 100 mW PEP, single-tone modulation per sideband
	F2 remainder of the day	
4	F3 first 4 hours of manned schedule	RATT (A7j) 100 mW average power keyed from space to mark at an 80 baud rate
	F4 remainder of the day	

1/ The same sequence shall be used on the next 4 days, except that the sequence of frequencies shall be reversed, F4 and F3 on the fifth day, F2 and F1 on the sixth day, F4 and F3 on the seventh day. The schedule shown in TABLE IV shall be repeated as necessary.

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4.6.5.2 Unmanned transmitting schedule. The mode of operation during the unmanned transmitting period shall be ISB (Fg), using TADIL A input (NET TEST) or modulated tones to both USB and LSB. Output shall be 100 mW PEP into the 50-ohm dummy load. The test frequency to be used during each unmanned period shall be the last frequency utilized during the period (see 4.6.5.1).

4.7 Maintainability demonstration. The maintainability demonstration shall be in accordance with 4.7.1 through 4.7.7.

4.7.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organization level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment MTTR requirement is fullfilled. The maintainability demonstration shall be performed just once during the first three months production.

4.7.1.1 Demonstration conditions. The equipment maintainability requirement shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCB's) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. One hundred candidate corrective maintenance faults shall be determined in accordance with Appendix A of MIL-STD-471. The information specified in a through c shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity, or its authorized representative, shall use the candidate tasks as a guide to determine a sample of 50 tasks for the demonstration.

4.7.2 Accept or reject criteria. Accept or reject criteria shall be as specified in 4.7.2.1.

4.7.2.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as indicated in TABLE V.

TABLE V. Acceptance criteria.

	MTTR	M_{maxct}
Acceptance level	20	0
Sample size	50	50

Acceptance shall occur when the number of observed corrective maintanance task times which exceed the required value of each specified index, MTTR or M_{maxct}, is less than or equal to that specified in the TABLE V corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and determined as greater than or less than each index. The actual value shall also be determined. An accept decision can only be made when an accept decision is made for both MTTR and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision. More than one erroneous failure indication shall constitute failure of the demonstration.

4.7.3 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.7.4 Technicians. The Government reserves the right to provide and select technicians to perform the maintainability demonstration.

4.7.5 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation to be delivered with the equipment.

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4.7.6 Rejection. Failure to conform to any of the requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the contracting officer shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies and
b. Upon approval of an approach, repeat the demonstration until an accept decision is reached.

4.7.7 Acceptance. The maintainability demonstration shall demonstrate that the equipment conforms to the accept criteria prior to delivery.

4.8 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for general purpose, CW, RATT and data communications, and may be used for surface ship, submarine, and fixed shore installation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify:

a. Title, number, and date of this specification
b. GFE, if required (see 3.2)
c. Interchangeability requirements, if required (see 3.3)
d. Quantity of accessory items required (see 3.4.1)
   1. Extender cables (see 3.15.1.33.1
   2. Extender cards (see 3.15.1.33.2
e. Rack mounting brackets, as required (see 3.15.1.2.1)
f. Number of first article samples to be submitted if other than specified in 4.3
g. When reinspected production control and environmental sample units may not be accepted (see 4.4.4)
h. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

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Paragraph	Data requirement	Applicable DID
4.5.8.15	Electromagnetic Interference Control Plan	DI-R-7061
4.5.8.15	Electromagnetic Interference Test Report	DI-R-7062
4.5.8.15	Electromagnetic Interference Test Plan	DI-R-7063

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a first article. The first article should consist of two unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 GFE. The contracting officer should arrange to furnish the equipment specified in 3.2.

6.5 Nomenclature. Nomenclature referred to herein assigned to the GFE units may be subject to change on any contract resulting in equipment produced in accordance with this specification.

6.6 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.7 Definitions. Definitions of terms used in this specification are given in 6.7.1 through 6.7.7.

6.7.1 Local input. The term local input in this specification indicates that control and modulating functions will be provided by the handset (or other appropriate device connected to the handset receptacle on the front panel) or by the hand key on the front panel.

6.7.2 Remote inputs. The term remote inputs in this specification indicates that control and modulating functions will be provided at the rear of the equipment case.

6.7.3 Standard two-tone audio input. Standard two-tone audio input is defined as two equal amplitude audio signals, 1000 Hz and 1625 Hz, with a level of 150 mV per tone supplied from a 600-ohm source.

6.7.4 Hum signals. Hum signals are any output products resulting from modulation at the primary power line frequency or power supply conversion frequencies or harmonics of these frequencies.

6.7.5 Standard single-tone input. Standard single-tone audio input is defined as a 1000 Hz tone, with a level of 150 mV at the remote input terminals.

6.7.6 Teletype input. Standard teletype input shall be 60 mA DC for the mark signal and 0 mA for space. The loop current source impedance shall be 1000-ohms.

6.7.7 Failure. Failure is any malfunction or parameter deviation that prevents the equipment from performing within the specified operational requirements.

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6.8 Maintenance concept. The equipment is part of a family of high frequency communication equipments which are designed to minimize equipment down-time through the use of plug-in assemblies. Combined with the maintenance complexity resulting from high density packaging, this results in a maintenance program entailing both organizational and depot level maintenance. Capabilities and limitations at those maintenance levels are as specified in a and b:

a. Organizational

1. Fault isolation, removal and replacement of discrete circuit components such as tubes, transistors, coils and capacitors (other than those contained in specified depot repairable modules); replacement of meters, indicator lamps, fuses; tuning and alignment.

2. Fault location and isolation of plug-in assemblies and piece parts, removal and replacement with use of required test equipment, and diagnostic and repair procedures, unless plug-in assembly complexity reaches or exceeds integrated circuitry.

b. Depot. Designated Naval shipyards and repair facilities will be required to perform any maintenance function required on piece-part component or end item including emergency manufacture of nonavailable materials, provide technical assistance to lower levels, and repair and restore the assemblies specified in 1 through 6:

1. Mode selector
2. IF amplifier
3. RF amplifier
4. Frequency standard
5. Translator and synthesizer
6. RATT generator

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY - EC

(Project No. 5820-N808)

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5. PROBLEM AREAS

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c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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